

CampusAssetIQ — Developer Project Overview

A Clear & Practical Walkthrough of What We Built for Swaminarayan University & PSM Hospital

To: IT Head of Department (HOD) & Team	Status: LIVE ONLINE 24/7 (Singapore Cloud)
From: IT Development / Engineering Desk	Live Website: https://it-support-6np1.onrender.com
Project: Campus Hardware Tracking & QR Helpdesk	Future Subdomain: support.swaminarayanuniversity.ac.in

1. Why We Built This (The Real Problems We Solved)

Until now, our campus and hospital IT management had a lot of manual headaches:

The Old Problem	How CampusAssetIQ Solves It
Scattered Excel Sheets Hardware records were spread across different files, making it hard to know which computer is in which room.	Single Central Database Every single CPU, monitor, keyboard, mouse, and printer is in one live database with exact Building, Floor, and Room locations.
Endless Phone Calls from Doctors When a PC in ICU or OPD broke down, doctors had to call IT and try to explain what PC is having the issue.	5-Second Smartphone QR Scan Doctors simply point their phone camera at the PC sticker. It opens a reporting page with the PC ID already filled in. No login needed!
Confusing Duplicate Names Calling PCs by generic names like 'ICU-PC-1' created chaos because multiple rooms had similar labels.	Single Permanent Asset Tag One unique tag (e.g. PSM/IT/2F/C-201) represents that exact machine forever across all reports and stickers.
Manual Servicing / PMS Tracking Remembering vendor warranty expiries and quarterly maintenance schedules was completely manual.	Automated PMS & Breakdown Register Clear dashboard showing which machines are due for service, MTTR (repair time), and active breakdown tickets.

2. How It Works in Real Life (The Core Features)

A. ■ The 5-Second Mobile QR Complaint System

This is the biggest convenience feature for our hospital staff. Every computer gets a physical sticker with a scannable QR code. When an issue happens:

1. The nurse or doctor points their phone camera at the QR code on the machine.
2. It opens `/report/<asset_id>/` instantly. The system **already knows** the Asset Tag, room number, and assigned user.
3. The doctor chooses the issue (e.g. 'Display Flickering' or 'Network Drop'), types 2 words, and taps Submit.
4. The ticket appears instantly on the IT Admin Desk dashboard. **Zero phone calls, zero confusion!**

B. ■■ Instant Asset Tag & Sticker Printing

Whenever you add a new PC or printer in the Inventory page, our system automatically creates a clean, ready-to-print sticker card in the **Asset Tag Center**. We designed it so only the **vital physical information** is printed on the tag (Asset ID, Device Type, Assigned Custodian, Room, Serial Number, and QR Code). We intentionally kept technical clutter (like internal RAM chip speeds or MAC addresses) off the physical sticker so it looks clean and readable from 5 feet away!

C. ■ Location Explorer (Building → Floor → Room)

You can click on any floor (e.g. PSM Hospital → 2nd Floor → ICU or Trauma Ward) and see every device running in that specific room at a glance, including who is responsible for it and whether it is active or in maintenance.

D. ■ Equipment Breakdown & Maintenance (PMS) Registers

Keeps an honest log of repairs, which technician fixed it, how many hours it was down, and when the next quarterly or annual servicing is due from the vendor.

3. How It Is Built (In Simple Engineering Words)

Component	What We Used	Why This Is Great For Our Campus
Backend	Python & Django 6.1	Rock-solid, super fast page loading. No heavy complicated JavaScript frameworks that slow down older hospital PCs.
Database	PostgreSQL 16 (Cloud)	Enterprise relational database. 100% reliable, zero risk of file corruption, completely independent of our laptops.
Cloud Host	Render (Singapore Edge)	Located close to India (low ping). Runs 24 hours a day, 7 days a week even when all our office laptops are shut down.
Security & SSL	Cloudflare HTTPS	Gives us the official green security lock. Modern smartphone cameras require HTTPS to allow QR code scanning.
Code Updates	GitHub Auto-Deploy	Whenever we make an improvement in code and push it to GitHub, the live cloud website automatically updates itself in 60 seconds.

4. Storage & Capacity (Can It Handle Our Entire Campus?)

YES, easily! Here is the exact capacity math for our cloud setup:

- **Database Capacity:** 1 GB dedicated storage on PostgreSQL. Since each equipment and ticket record is clean text (~0.5 KB), 1 GB can hold **over 20,00,000 (20 Lakh) records!**
- **Real Campus Consumption:** Our university and hospital will create roughly ~10 MB of data per year (including 5,000 PCs and 10,000 repair tickets). That means our current setup gives us **50 to 100 YEARS of storage** before getting full.
- **Traffic Capacity:** 100 GB of free bandwidth per month, which easily covers **over 10,00,000 (10 Lakh) page views every month.**

5. Connecting Our Official Subdomain (Next 2-Minute Step)

Our site is already working 24/7 on: <https://it-support-6np1.onrender.com>. To make it open when someone types support.swaminarayanuniversity.ac.in, we just need our IT network admin to add **ONE standard CNAME record**:

Type:	CNAME
Host / Name:	support
Points to / Value:	it-support-6np1.onrender.com
Impact on main site:	ZERO RISK. Main university site (swaminarayanuniversity.ac.in) and emails stay 100% untouched.

6. Summary & How We Can Roll This Out

1. **Test the live link on your phone:** Open <https://it-support-6np1.onrender.com/report/> and see how smoothly the QR complaint page loads.
2. **Print a sample sheet of QR tags:** From /tag/, print 1 A4 sheet on our office printer and test scanning with any smartphone camera.
3. **Add the DNS CNAME:** Have our domain manager point support to it-support-6np1.onrender.com.

Submitted by: IT Systems Development Desk • Swaminarayan University & PSM Hospital